Agentic and Gen AI Design Pattern

PAIR- AI:
Motif library:

Gen AI:

- RAG doi 10.48550/arXiv.2312.10997
  - Iterative, recursive, adaptive

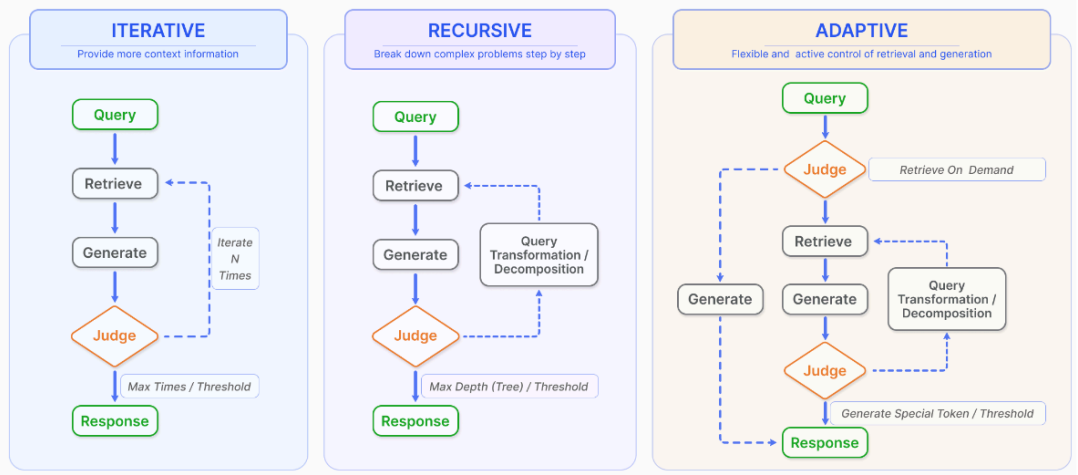


Fig. 5. In addition to the most common once retrieval, RAG also includes three types of retrieval augmentation processes. (left) Iterative retrieval involves alternating between retrieval and generation, allowing for richer and more targeted context from the knowledge base at each step. (Middle) Recursive retrieval involves gradually refining the user query and breaking down the problem into sub-problems, then continuously solving complex problems through retrieval and generation. (Right) Adaptive retrieval focuses on enabling the RAG system to autonomously determine whether external knowledge retrieval is necessary and when to stop retrieval and generation, often utilizing LLM-generated special tokens for control.

- Guard Rail doi 10.48550/arXiv.2109.01652

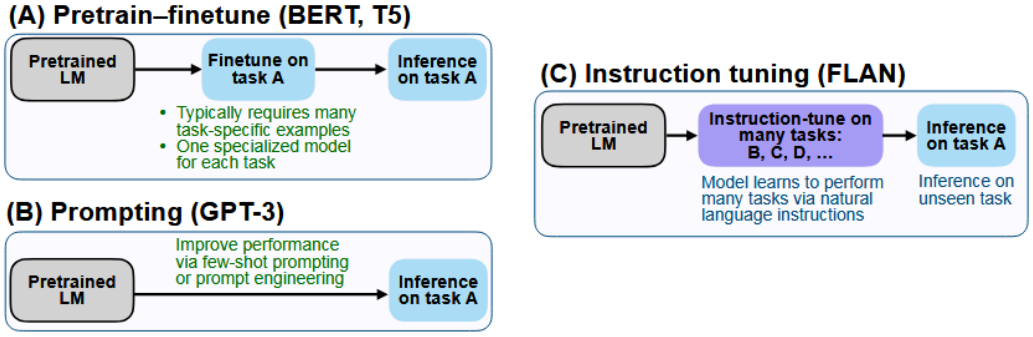
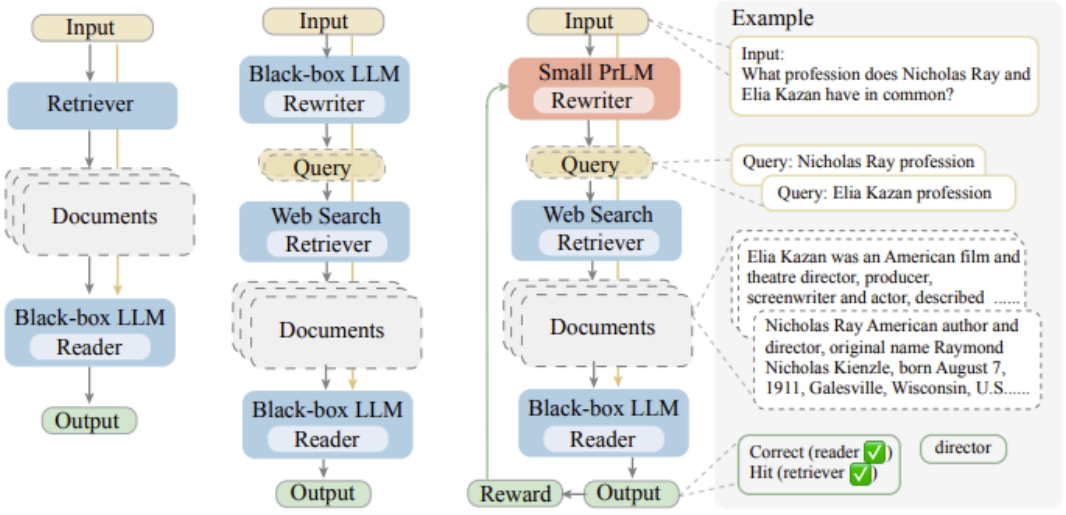


Figure 2: Comparing instruction tuning with pretrain–finetune and prompting.

- Context Bloat
  - Lost in the Middle: How Language Models Use Long Contexts, doi `10.1162/tacl_a_00638"`:
- Query-rewriting

  https://doi.org/10.48550/arXiv.2305.14283



(a) Retrieve-then-read (b)Rewrite-retrieve-read (c) Trainable rewrite-retrieve-read

re 1: Overview of our proposed pipeline. From left to right, we show (a) standard *retrieve-then-read* method, LM as a query rewriter for our *rewrite-retrieve-read* pipeline, and (c) our pipeline with a trainable rewriter.

- LLM based Information Retrieval

doi: https://doi.org/10.1007/s13222-025-00503-x

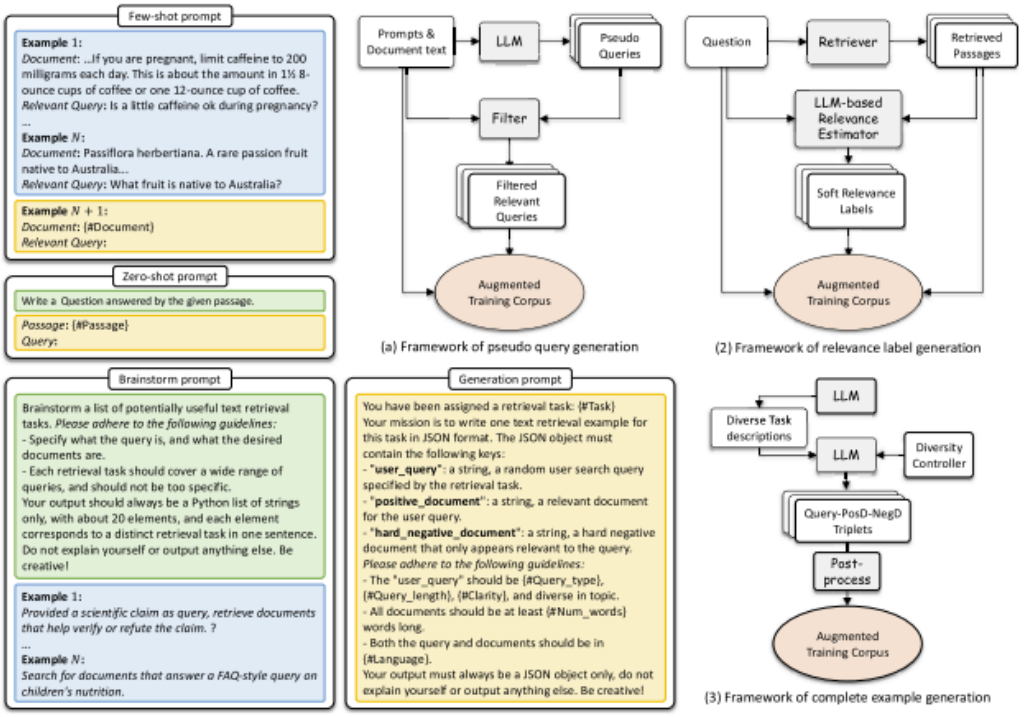


Fig. 4. Three typical frameworks for LLM-based data augmentation in the retrieval task (right), along with their prompt examples (left). Note that the methods of relevance label generation do not treat questions as inputs but regard their generation probabilities conditioned on the retrieved passages as soft relevance labels.

Agentic A I and Gen AI (Updated) doi (10.1016/j.jss.2024.112278) :

- Passive Goal Creator

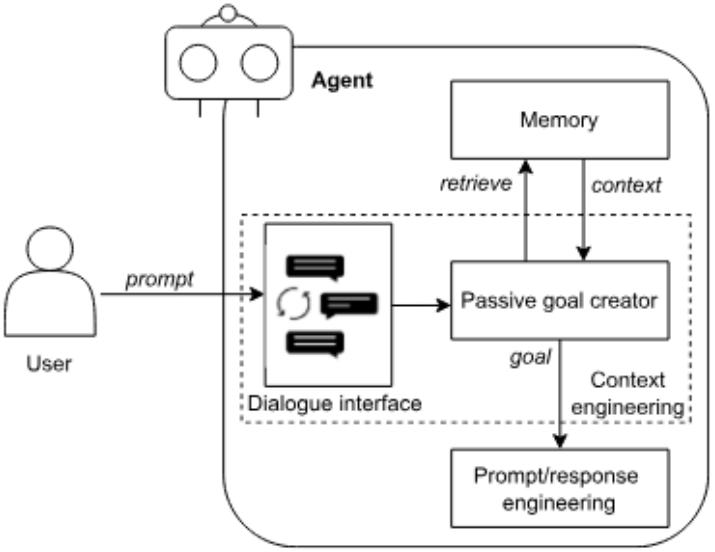


Fig. 3. Passive goal creator.

- Proactive goal creator

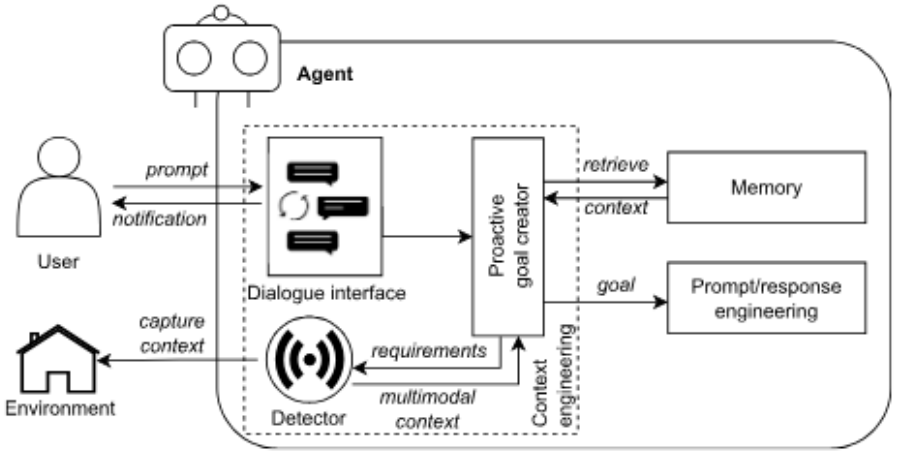


Fig. 4. Proactive goal creator.

- Prompt/respone optimiser

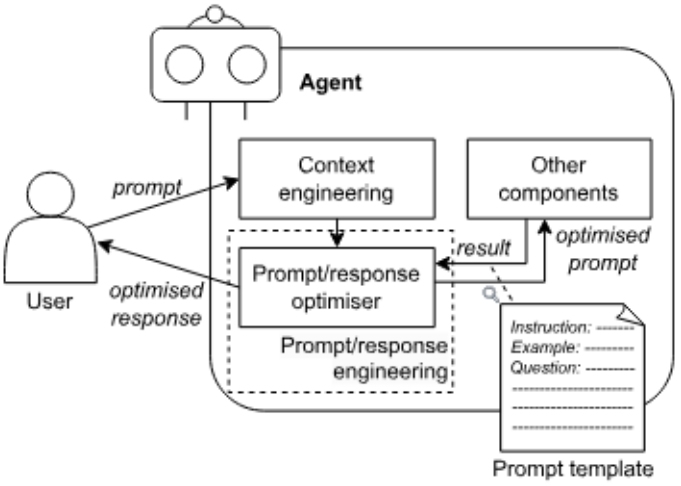


Fig. 5. Prompt/response optimiser.

- RAG

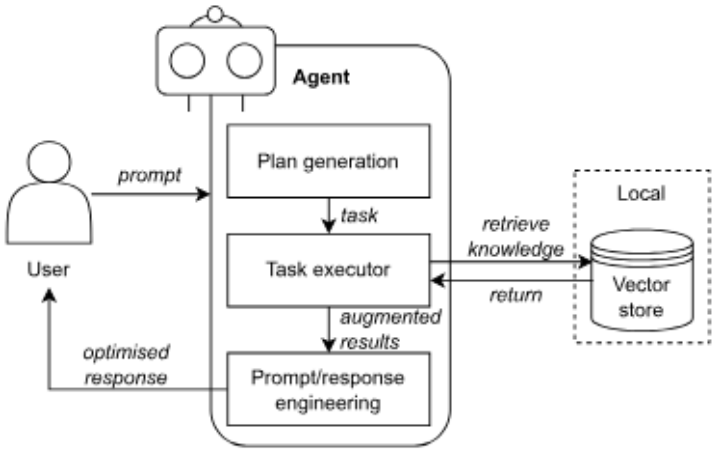


Fig. 6. Retrieval augmented generation.

- One-shot model querying

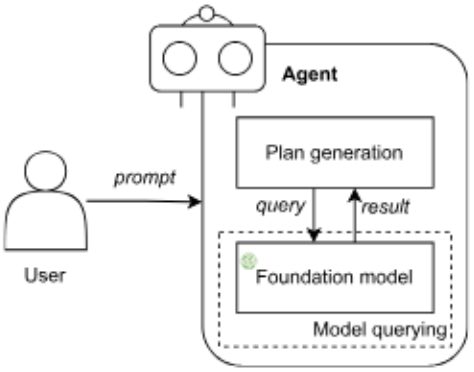


Fig. 7. One-shot model querying.

- Incremental model querying

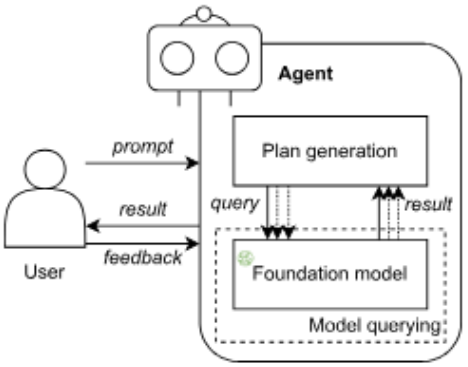


Fig. 8. Incremental model querying.

- Multi-path plan generator

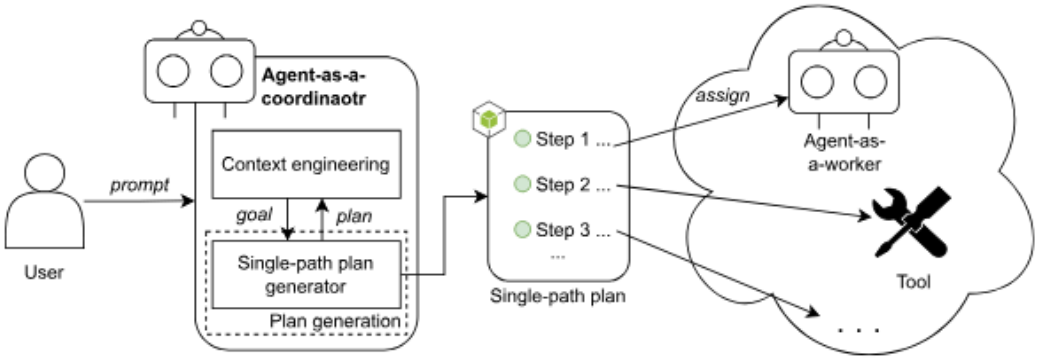


Fig. 9. Single-path plan generator.

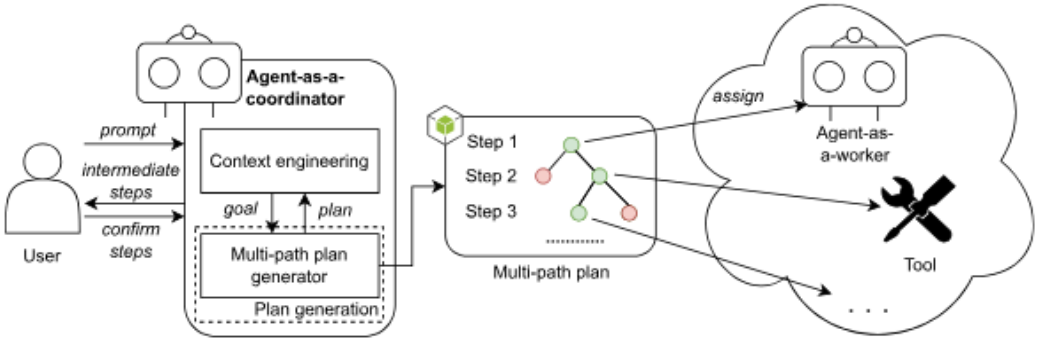


Fig. 10. Multi-path plan generator.

- Voting based Coorperation

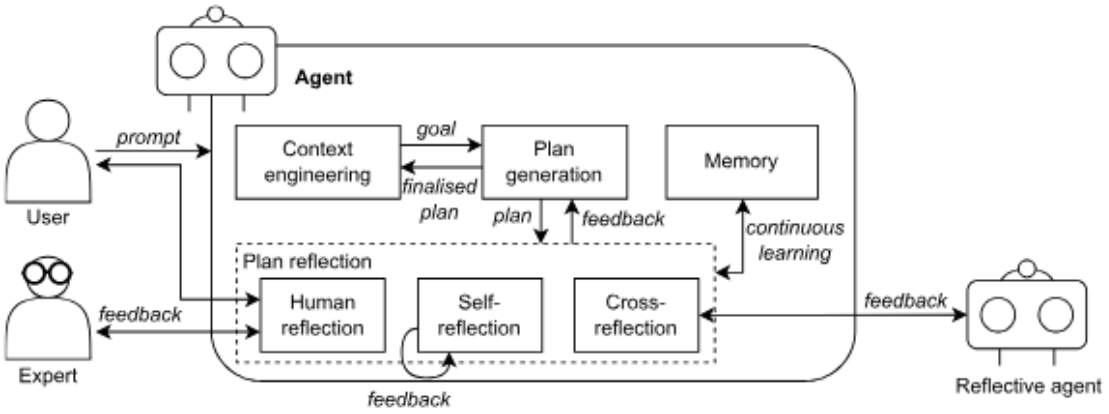


Fig. 11. Plan reflection pattern.

- Role based coorperation

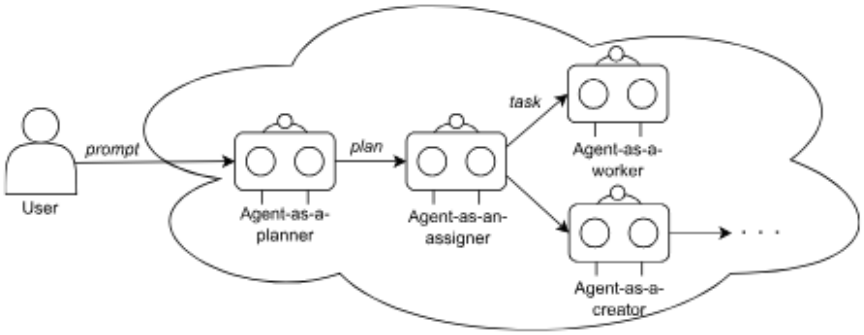


Fig. 13. Role-based cooperation.

- debate based coorperation

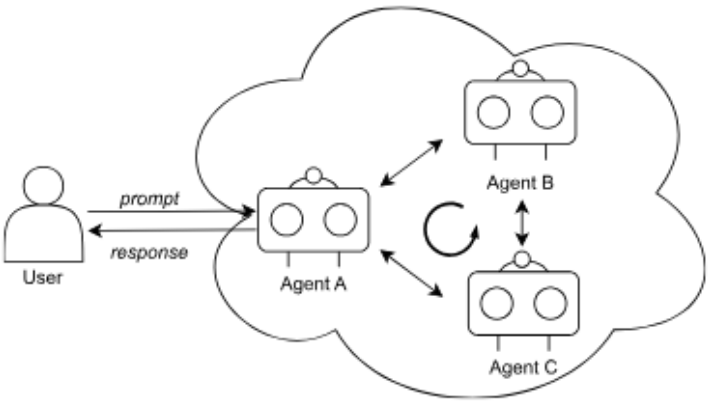


Fig. 14. Debate-based cooperation.

- multi modelg guardrail

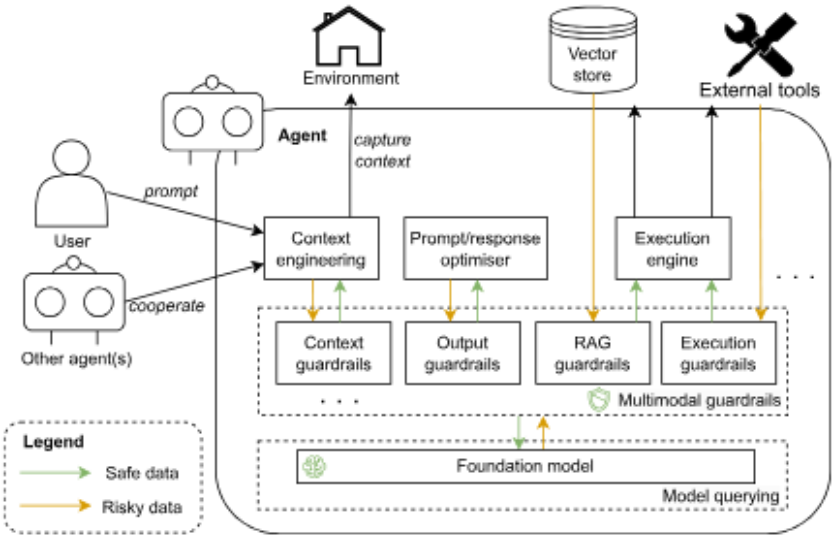


Fig. 15. Multimodal guardrails.

- tool agent registry

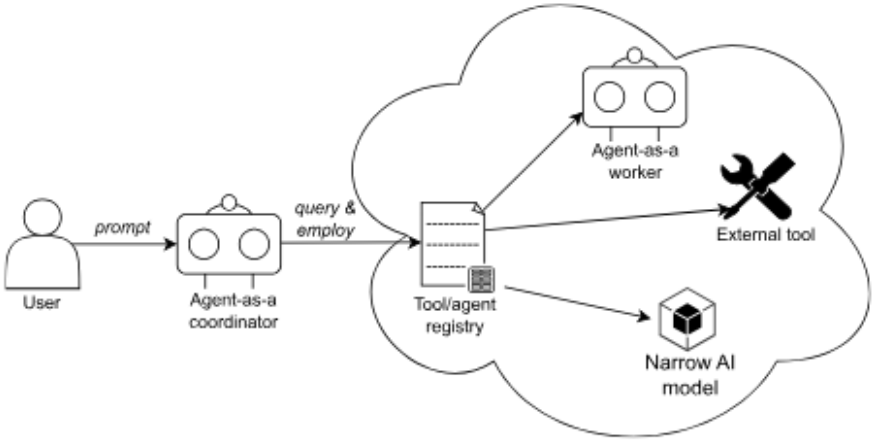


Fig. 16. Tool/agent registry.

- agent adapter

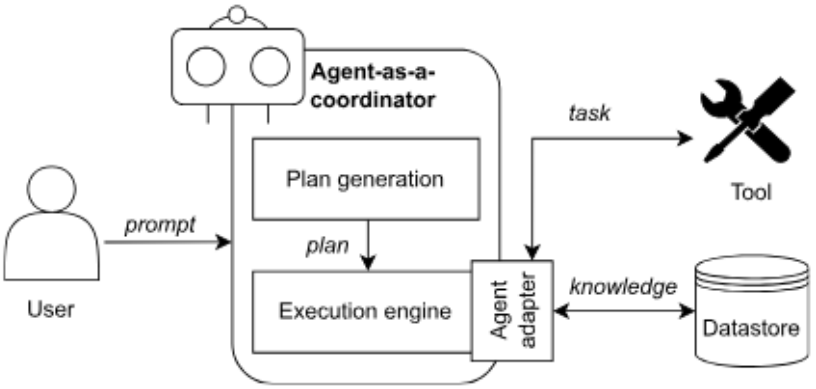


Fig. 17. Agent adapter.

- agent evaluator

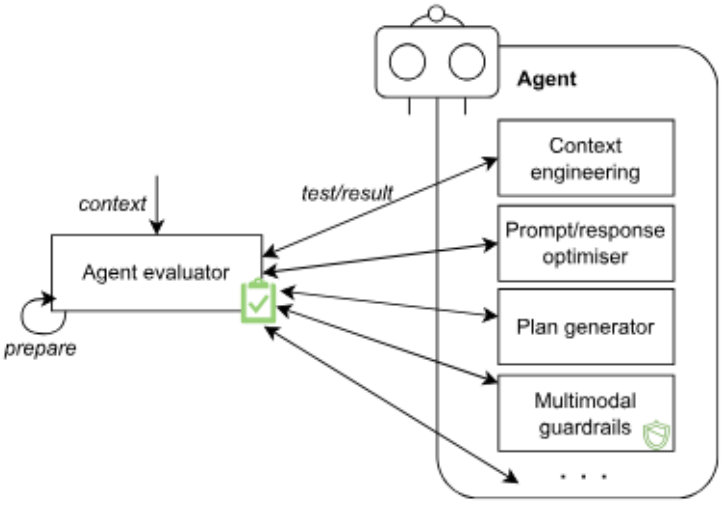


Fig. 18. Agent evaluator.